

6.7980: Topics in Multiagent Learning

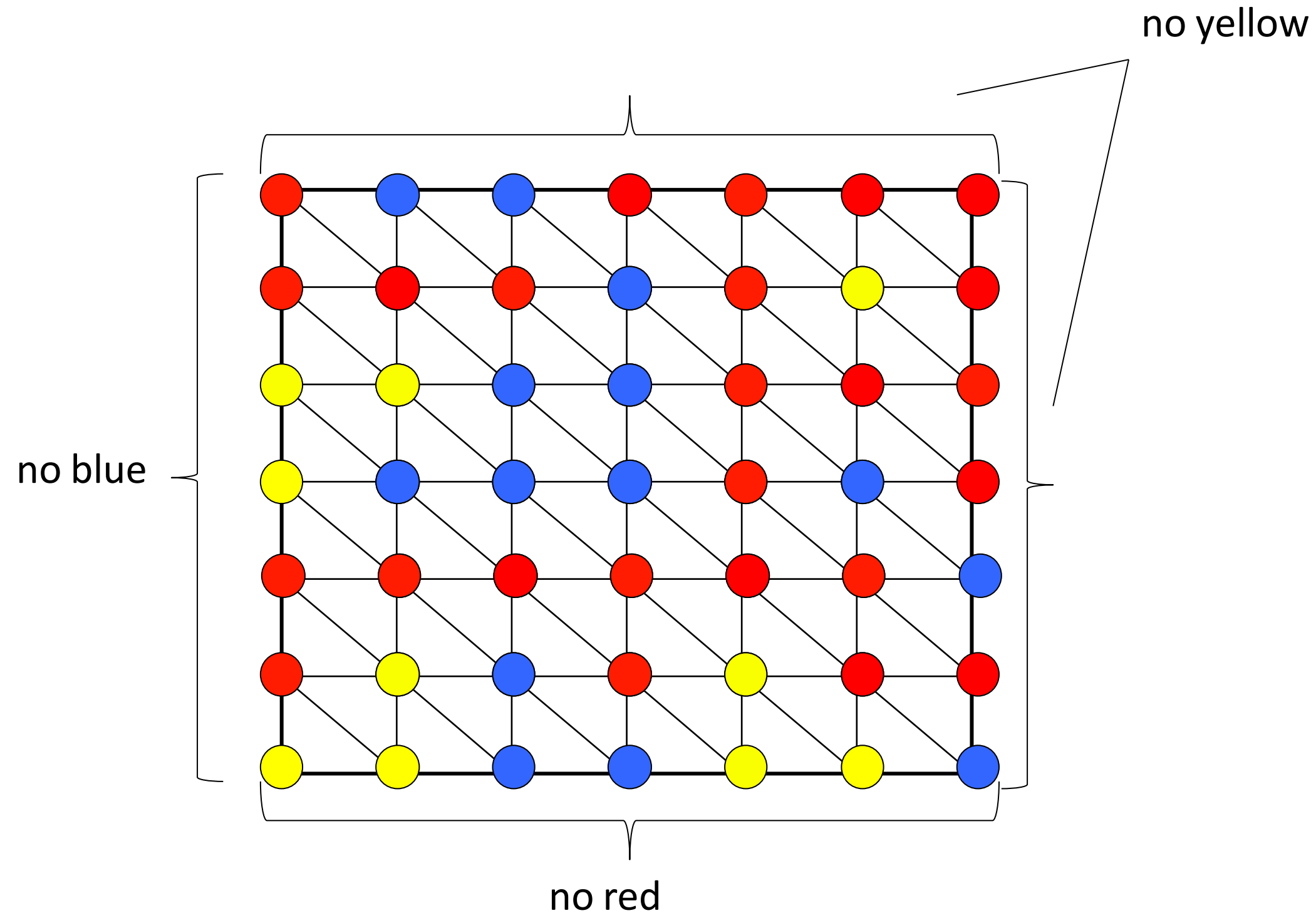
Lecture 3 – Prof. Daskalakis
Fall 2026



Refresher

- Last couple of times we stated and proved the theorems of Nash, Sperner, and Brouwer.
- [Nash'1950]: Every finite game has a Nash equilibrium.
- [Brouwer'1911]: Every continuous function $f: D \rightarrow D$ from a convex compact set D to itself has a fixed point $x^* = f(x^*)$.
- [Sperner'1928]: Every legal 3-coloring of a 2-d triangulated square has a tri-chromatic triangle.

Sperner's Lemma (in 2-d)



[Sperner 1928]: Color the boundary using three colors in a legal way. No matter how the internal nodes are colored, there exists a tri-chromatic triangle. In fact an odd number of those.

Refresher

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- In particular, we saw that:

Sperner Lemma $\Rightarrow$ Brouwer Theorem $\Rightarrow$ Nash Theorem

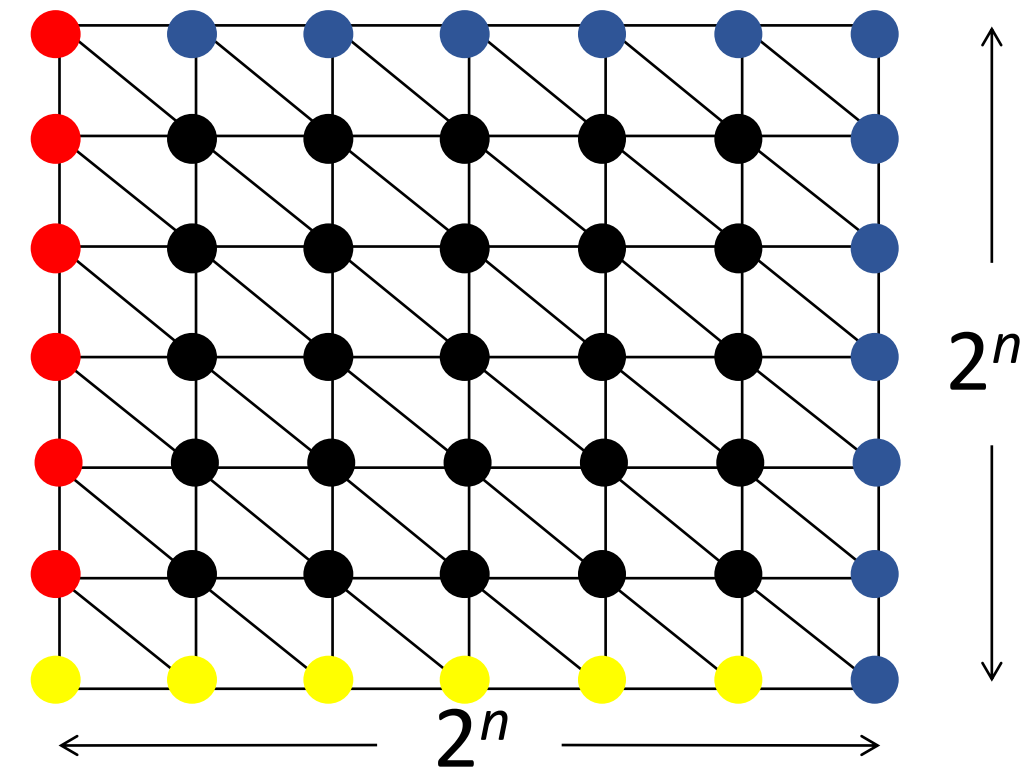
- Computational implications:
 - Finding Nash Equilibria *poly-time reduces* to Finding Brouwer Fixed points
 - Finding Brouwer Fixed Points *poly-time reduces* to Finding Sperner Triangles
 - The problem is that the resulting Sperner instance from these poly-time reductions is enormous..

Computational Problem SPERNER

INPUT:

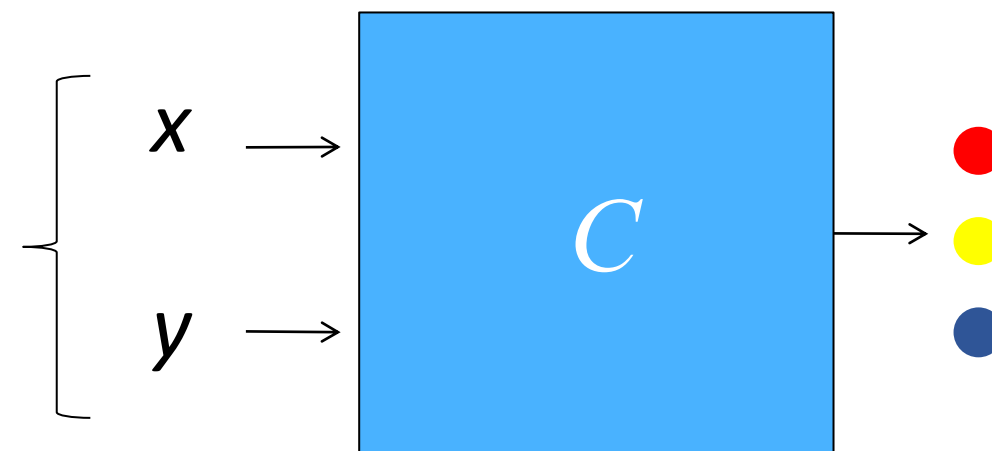
(i) n : specifies the size of a grid

(grid never written down!)



(ii) Imagine boundary has standard coloring shown above, while colors of internal vertices are given by a circuit:

input: the
coordinates
of a point
(n bits each)



OUTPUT: A tri-chromatic triangle

*exists because boundary coloring satisfies Sperner lemma constraints
but doing walk through grid to find one may take exponential time in n*

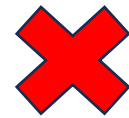
So what is the complexity of NASH in general games?

- No polynomial-time algorithm has been found for NASH in ~75 years, for SPERNER in ~100 years, and for BROUWER in ~115 years respectively
- All these problems are all in NP, but are likely not NP-Hard
 - because if they were, $\text{co-NP} = \text{NP}$
 - because of the guaranteed existence of solutions
- So what is their complexity?
 - So far, we have seen the polynomial-time reductions:
$$\text{NASH} \leq_p \text{BROUWER} \leq_p \text{SPERNER}$$
 - Later this semester we will see that:
$$\text{NASH} =_p \text{BROUWER} =_p \text{SPERNER} \text{ [Daskalakis-Goldberg-Papadimitriou'06]}$$
 - and, in particular, they are all PPAD-complete, where PPAD is a complexity class that lies between P and NP
 - Unless $P = \text{PPAD}$, there is no polynomial-time algorithm for these problems

Score Card

Computational
Complexity

General Games



Algebraic and Topological Complexity

Are Nash equilibrium probabilities even rational?

- It was already shown by Nash'51 that there exist 3-player games whose only Nash equilibria use irrational numbers in the probabilities.
- Here is a simple example by [Nau, Canovas and Hansen'04]

	<i>left</i>	<i>right</i>		<i>left</i>	<i>right</i>
<i>top</i>	3, 0, 2	0, 2, 0	<i>top</i>	1, 0, 0	0, 1, 0
<i>bottom</i>	0, 1, 0	1, 0, 0	<i>bottom</i>	0, 3, 0	2, 0, 3
	<i>(action X)</i>			<i>(action Y)</i>	

whose unique Nash equilibrium is

$$\left(\frac{53}{46} - \frac{\sqrt{601}}{46}, -\frac{13}{24} + \frac{\sqrt{601}}{24}, -\frac{23}{4} + \frac{\sqrt{601}}{4} \right) \approx (0.619, 0.480, 0.379)$$

- Even worse, [Datta '03] showed that for any real algebraic variety, one can come up with some three-player game whose set of totally mixed Nash equilibria is isomorphic to that variety.

Score Card

	Computational Complexity	Algebraic Complexity	Topological Complexity
General Games	✗	✗	✗

How about two-player games?

- A 2-player can be summarized by two matrices $(R, C)_{m \times n}$
 - rows : indexed by pure strategies of “row player”
 - columns : indexed by pure strategies of “column player”
- Mixed strategy for row player: $x \in \Delta^m$
- Mixed strategy for column player: $y \in \Delta^n$
- Expected utility of row player: $u_{\text{row}}(x, y) = x^T R y = \sum_{ij} R_{ij} x_i y_j$
- Expected utility of column player: $u_{\text{column}}(x, y) = x^T C y = \sum_{ij} C_{ij} x_i y_j$
- (x, y) is Nash equilibrium iff
$$\begin{aligned} \forall x': x^T R y &\geq x'^T R y \\ \forall y': x^T C y &\geq x^T C y' \end{aligned}$$

Two-player *Zero-Sum* games

Minimax Theorem [von Neumann'28]: Consider a two-player game zero-sum game $(R, C)_{m \times n}$ i.e. $R + C = 0$. Then $\min_{x \in \Delta^m} \max_{y \in \Delta^n} x^T C y = \max_{y \in \Delta^n} \min_{x \in \Delta^m} x^T C y$ (*)

Interpretation:

- For any value v :
 - $\text{RHS} \leq v$ means " $\forall y, \exists x$ s.t. $x^T C y \leq v$ "
 - $\text{LHS} \leq v$ means " $\exists x, \forall y$ s.t. $x^T C y \leq v$ "
- (*) says: "If $\forall y, \exists x$ s.t. $x^T C y \leq v \Rightarrow \exists x, \forall y$ s.t. $x^T C y \leq v$ "
- If x^* is argmin of LHS, y^* argmax of RHS, v^* optimal value of (*), then (x^*, y^*) is a *Nash equilibrium*, i.e. if *min* and *max* adopt x^* and y^* then (i) *min* pays v^* to *max* and (ii) no player can improve by unilaterally deviating
- why? Because
 - under (x^*, y^*) *min* pays *max* at most v^* (since v^* optimum of LHS and x^* is argmin)
 - under (x^*, y^*) *max* receives from *min* at least v^* (since v^* optimum of RHS and y^* is argmax)
 - by the above two: under (x^*, y^*) *min* pays exactly v^* to *max*, hence (i) is proven
 - to prove (ii), suppose $\exists x$ that is a better response for *min* to y^* i.e. $x^T C y^* < x^{*T} C y^* = v^*$
 - the existence of such x violates the fact that the optimum of RHS is v^* and y^* is an argmax for RHS
 - similarly the existence of a better response to x^* by *max* violates that the optimum of LHS is v^* and x^* is an argmin for the LHS

Two-player *Zero-Sum* games

Minimax Theorem [von Neumann'28]: Consider a two-player game zero-sum game $(R, C)_{m \times n}$ i.e. $R = -C$. Then $\min_{x \in \Delta^m} \max_{y \in \Delta^n} x^T C y = \max_{y \in \Delta^n} \min_{x \in \Delta^m} x^T C y$ (*)

- **Original Proof:** used Brouwer's fixed point theorem.
- Here we'll do a proof using Strong Linear Programming duality
- Proof:

LHS:

$$\begin{aligned} LP1: \quad & \min z \\ & x^T C e_j \leq z, \forall j \\ & x \in \Delta^m \end{aligned}$$

RHS:

$$\begin{aligned} LP2: \quad & \max w \\ & e_i^T C y \geq w, \forall i \\ & y \in \Delta^n \end{aligned}$$

LP1 and LP2 are duals!
Strong LP duality: LP1=LP2

Two-player *Zero-Sum* games

Minimax Theorem [von Neumann'28]: Consider a two-player game zero-sum game $(R, C)_{m \times n}$ i.e. $R = -C$. Then $\min_{x \in \Delta^m} \max_{y \in \Delta^n} x^T C y = \max_{y \in \Delta^n} \min_{x \in \Delta^m} x^T C y$ (*)

Corollary: Every finite two-player zero-sum game has a Nash equilibrium.

Original Proof: used Brouwer's fixed point theorem.

Our Proof: used strong linear programming duality

Corollary: The set of Nash equilibria is a Cartesian product of nonempty, convex, compact sets.

Corollary: The set of Nash equilibria can be computed in polynomial-time.

Connection to mathematical programming:

- **[von Neumann-Dantzig'47, Adler'13, Brooks-Reny'21]:** minimax eq computation $\Leftrightarrow$ Linear Programming

von Neumann and Dantzig



[picture from Game Theory Alive, by Anna Karlin and Yuval Peres]

- On October 3, 1947, I visited him (von Neumann) for the first time at the Institute for Advanced Study at Princeton.
- I remember trying to describe to von Neumann, as I would to an ordinary mortal, the Air Force problem. I began with the formulation of the linear programming model in terms of activities and items, etc.
- Von Neumann did something which I believe was uncharacteristic of him. “Get to the point,” he said impatiently. Having at times a somewhat low kindling-point, I said to myself “O.K., if he wants a quicky, then that’s what he will get.”
- In under one minute I slapped the geometric and algebraic version of the problem on the blackboard. Von Neumann stood up and said “Oh that!” Then for the next hour and a half, he proceeded to give me a lecture on the mathematical theory of linear programs.
- At one point seeing me sitting there with my eyes popping and my mouth open (after I had searched the literature and found nothing), von Neumann said: “I don’t want you to think I am pulling all this out of my sleeve at the spur of the moment like a magician. I have just recently completed a book with Oskar Morgenstern on the theory of games. What I am doing is conjecturing that the two problems are equivalent. The theory that I am outlining for your problem is an analogue to the one we have developed for games.” Thus I learned about Farkas’ Lemma, and about duality for the first time.

Score Card

	Computational Complexity	Algebraic Complexity	Topological Complexity
General Games			
Two-Player Zero-Sum Games			

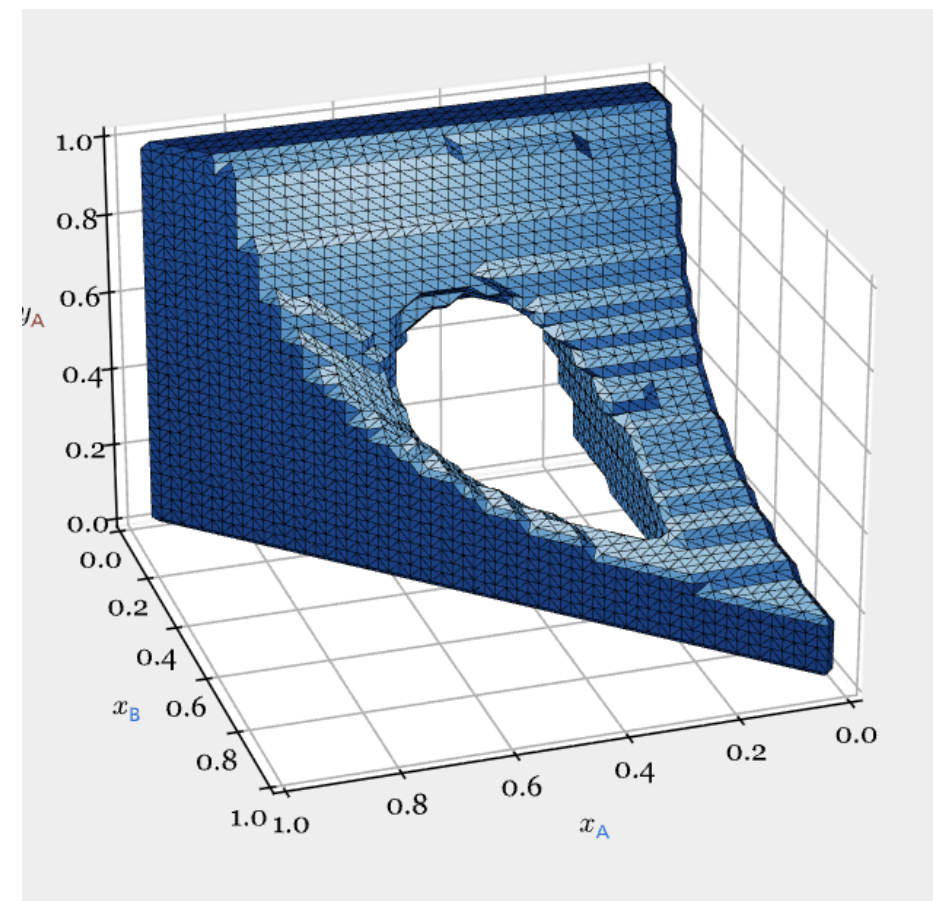
How about two-player general-sum games?

Good news: There exist Nash equilibria in rational probabilities!

Bad news 1: The of Nash equilibria in a game can be quite complex, e.g. the set of Nash equilibria might have holes.

	A	B	C
A	1, 1	0, -1	-1, 1
B	-1, 0	0, 0	-1, 0
C	1, -1	0, -1	-2, -2

[Kohlberg-Mertens'86]



projection of the set of all 0.27-approximate Nash equilibria of this game [Milionis, J., Papadimitriou, C., Piliouras, G., & Spendlove, K., PNAS 2023]

Bad news 2: Computing them is PPAD-complete as well, i.e. exactly as hard as solving SPERNER, BROUWER, Multi-Player NASH

Score Card

	Computational Complexity	Algebraic Complexity	Topological Complexity
General Games	✗	✗	✗
Two-Player Zero-Sum Games	✓	✓	✓
Two-Player General-Sum Games	✗	✓	✗